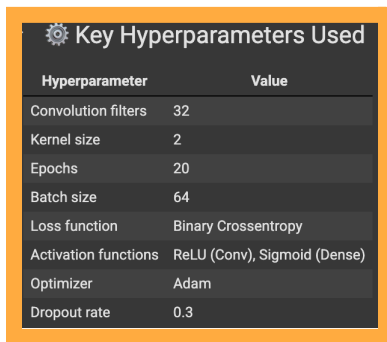


Introduction

We approached the classification task using a **convolutional neural network (CNN)** architecture, chosen for its suitability in capturing **spatial patterns** across weather stations. Each data sample represented a single timestep across 15 stations, with 9 features per station — making this a **multi-label classification problem** with 15 binary outputs per sample. Given the spatial nature of the data, we determined that a CNN would be more appropriate than a recurrent network, which is typically more effective for temporal sequences.

We conducted **three structured iterations**, adjusting the architecture and training configuration based on validation dynamics, test performance, and per-station metrics.

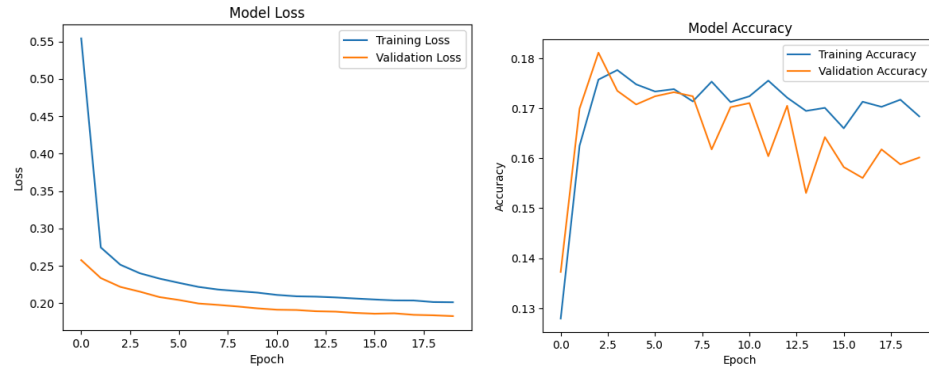
Model 1: Baseline CNN



Key Hyperparameters Used	
Hyperparameter	Value
Convolution filters	32
Kernel size	2
Epochs	20
Batch size	64
Loss function	Binary Crossentropy
Activation functions	ReLU (Conv), Sigmoid (Dense)
Optimizer	Adam
Dropout rate	0.3

We trained the model for **20 epochs**, using **binary crossentropy** as the loss function and **Adam** as the optimizer. This version produced a **test loss of 0.1827** and a **global F1-score of 95% on the majority class**, and **81% on the minority class (“pleasant” days)**. Per-station F1 scores ranged from **91% (Madrid)** to **53% (Valentia)**, with **Sonnblick scoring 0** due to no positive cases in the test set.

However, performance curves suggested room for improvement:



- **Validation accuracy peaked early** (around epoch 2) and steadily declined
- **Training accuracy remained higher**, a classic sign of **overfitting**
- Meanwhile, **both training and validation loss decreased in parallel**, which indicated the model was still learning structure — but perhaps under constraints that limited further generalization.

While dropout was intended to reduce overfitting, a **rate of 0.3** may have been **too aggressive** early in training, potentially stalling learning. This prompted a second run with a more balanced setup.

Model 2: Relaxed Regularization and Early Stopping

Key Hyperparameters Used	
Hyperparameter	Value
Convolution filters	32
Kernel size	2
Epochs (max)	15
Batch size	64
Loss function	Binary Crossentropy
Activation functions	ReLU (Conv), Sigmoid (Dense)
Optimizer	Adam
Dropout rate	0.1
EarlyStopping	Patience = 3, Monitor = val_loss, Restore Best = True

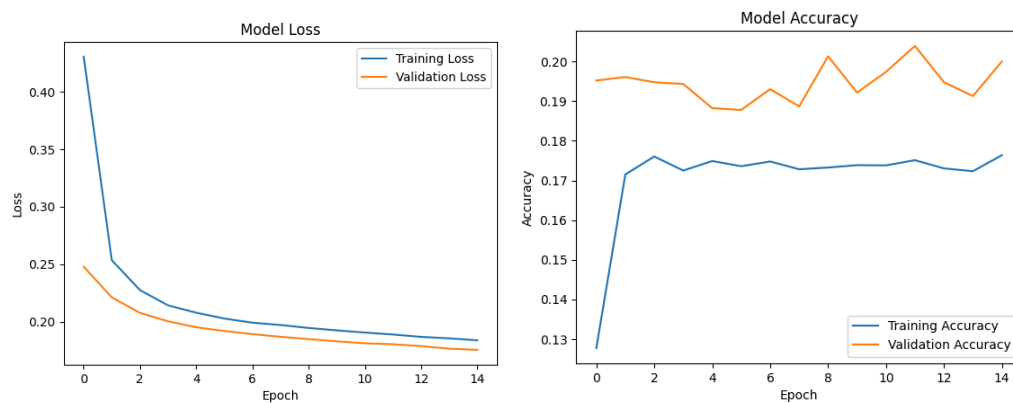
In the second model, we made **two key adjustments**:

- Reduced **dropout rate from 0.3 to 0.1**, allowing the model to retain more representational capacity
- Introduced **early stopping**, monitoring validation loss to avoid overfitting in later epochs
- Reduced the max epoch count from 20 to **15**, anticipating faster convergence

These changes aimed to create a **less constrained learning environment** while keeping overfitting in check via early stopping.

Performance changes were subtle but meaningful:

- **Test loss decreased** slightly to **0.1757**; **Test accuracy increased** to 0.1974 (a small, but more noticeable increase from ~0.16)
- Global F1 scores stayed stable at **95% (majority)** and **81% (minority)**
- However, **minority class precision improved slightly**, with a small trade-off in recall; for most weather stations, the change in model was marked by an increase in precision vs. recall. The reshuffling of per-station performance — with precision generally increasing — suggests that the model was learning more confidently, if slightly more conservatively.



- Validation loss remained consistently below training loss, with the two tracking more closely rather than staying in parallel; validation accuracy stayed consistently high above testing accuracy.

Model 3: Expanded Architecture and Finer Learning Adjustments

Key Hyperparameters Used (Model #3)	
Hyperparameter	Value
Conv layers	2 (both with 32 filters)
Kernel size	3
Dropout	0.15
Epochs	15
Batch size	32
Loss	Binary Crossentropy
Optimizer	Adam
Activation functions	ReLU (Conv), Sigmoid (Dense)

In our third and final run, we focused on deepening the model's ability to extract **hierarchical spatial features**. To that end, we made the following changes:

- **Added a second Conv1D layer** with the same number of filters (32), to allow for

deeper pattern extraction across stations

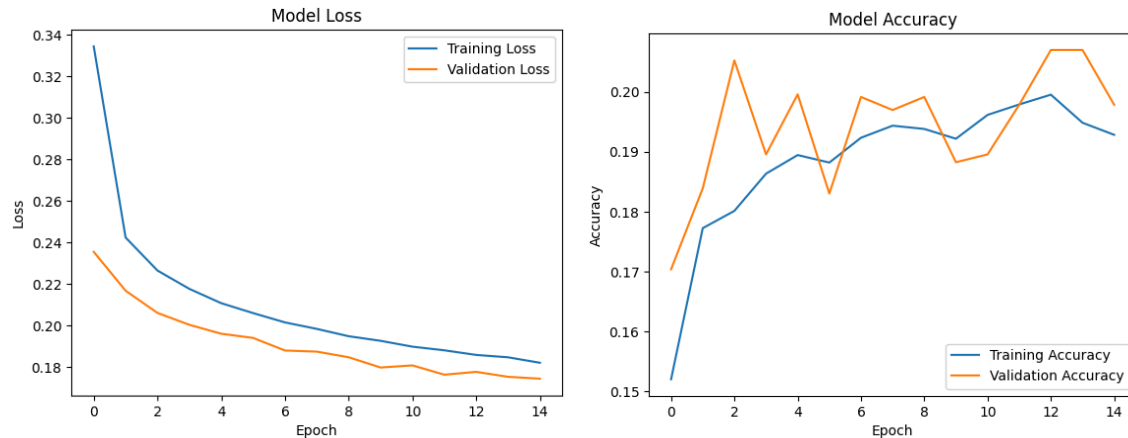
- **Increased the kernel size** from 2 to 3, giving the model a slightly wider receptive field — effectively letting it observe broader station-to-station patterns
- **Reduced batch size** from 64 to 32, enabling more frequent weight updates, which can sometimes improve generalization, particularly on smaller datasets or noisier tasks

This model achieved the **lowest test loss of all three (0.1740)** — only a marginal drop from Model 2's 0.1757, but directionally consistent. Test accuracy also rose to **0.200**, the highest yet, though again this reflects Keras' strict metric requiring all 15 station outputs to be correct. Importantly, the **global per-label accuracy remained at 92%**, consistent across all three models — a stable baseline.

The overall F1 scores held steady at **95% for the majority class** and **81% for the minority ("pleasant") class**, but as in Model 2, we observed a further **increase in minority precision** paired with a **slight decrease in recall**. This suggests a model that has become more **cautious** — favoring fewer false positives, even at the cost of missing some true positives. For certain real-world applications (like travel planning), this shift may be desirable.



However, not all changes were uniformly beneficial: **Valentia's F1 and precision dropped noticeably**, suggesting the model may have grown more risk-averse in low-support scenarios. This reinforces the case for exploring **class weighting or focal loss** in future work to better support underrepresented stations.



In terms of training dynamics:

- **Both training and validation accuracy rose steadily across epochs**, without the early collapse or plateauing seen in prior models
- That said, **accuracy curves remained volatile**, especially on the validation set — likely due to the smaller batch size and increased model complexity

Despite this, the aggregate improvement in validation performance and precision — without sacrificing minority class performance — suggests **Model 3 achieved the best balance** of generalization and confidence.

Conclusion

Across all three CNN models, performance remained strong and consistent: **92% global accuracy**, with **F1 scores of 95% (majority)** and **81% (minority)**. Each iteration refined how the model balanced caution and generalization.

Adding a second **Conv1D** layer and increasing kernel size in Model 3 gave the **lowest test loss** and **highest strict accuracy** (0.200), while progressively **increasing precision** and **slightly lowering recall** for the minority class. This trend suggests our models became more conservative — good for avoiding false positives, but at a cost to sensitivity.

The final drop in F1 for low-support stations (e.g. Valentia) highlights the limits of equal class treatment. For future work, **class weighting** or **custom loss adjustments** will likely be essential to improve fairness and performance on underrepresented stations.